

Addendum: Viscous Stable Cosserat Rods

Jerry Hsu
jerry.hsu.research@gmail.com

March, 2026

Abstract

We extend the Stable Cosserat Rods framework [1] with a Kelvin–Voigt style strain rate damping term on the stretching and shearing strain. The damping is formulated in the local material frame to avoid penalizing rigid body rotation. By exploiting the structural similarity to the existing stretching/shearing energy, the extension integrates into the solver with minimal modification: the material axis e_3 is replaced by a generalized axis $\boldsymbol{\eta}$ that encodes both elastic and viscous contributions.

1 Background

We briefly review the relevant parts of the Stable Cosserat Rods formulation. A discretized Cosserat rod stores positions $\mathbf{x}_i$ on vertices and unit quaternion orientations q_i on segments. The stretching and shearing constraint on segment i is

$$C_i^{ss} = \frac{\mathbf{x}_{i+1} - \mathbf{x}_i}{l_i} - d_{i,3}, \quad d_{i,3} = q_i e_3 \bar{q}_i, \quad (1)$$

with rest length l_i . The bending and twisting constraint between segments i and $i + 1$ is

$$C_i^{bt} = \bar{q}_i q_{i+1} - \phi_i q_i^0, \quad (2)$$

where q_i^0 is the rest relative orientation and ϕ_i selects the nearest quaternionic pole. The total elastic energy is

$$E^{total} = \sum_i \frac{k_i^{ss}}{2} |C_i^{ss}|^2 + \sum_i \frac{k_i^{bt}}{2} |C_i^{bt}|^2. \quad (3)$$

Under the quasi-static orientation assumption ($J = 0$), the orientation solve reduces to a local Gauss–Seidel relaxation. For each segment i , the equilibrium condition with Lagrange multiplier λ enforcing $|q_i| = 1$ is

$$\mathbf{v} q_i e_3 + \mathbf{b} - \lambda q_i = 0, \quad (4)$$

where the stretching contribution $\mathbf{v}$ and bending contribution $\mathbf{b}$ are

$$\mathbf{v} = -\frac{2k_i^{ss}}{l_i} (\mathbf{x}_{i+1} - \mathbf{x}_i), \quad \mathbf{b} = \sum_{C^{bt} \ni i} (\text{bending terms}). \quad (5)$$

The closed-form solution is obtained by right-multiplying eq. (4) by e_3 , using $e_3^2 = -1$ to isolate $q_i e_3$, and substituting back:

$$q_i(\lambda) = \frac{\mathbf{v} \mathbf{b} e_3 + \lambda \mathbf{b}}{\lambda^2 - |\mathbf{v}|^2}, \quad (6)$$

with approximate $\lambda \approx |\mathbf{v}| + |\mathbf{b}|$.

2 Damping Energy

We define the damping potential as the squared magnitude of the time rate of change of the local stretching and shearing strain. The local-space strain on segment i is

$$\Gamma_i^{loc} = \bar{q}_i \frac{\mathbf{x}_{i+1} - \mathbf{x}_i}{l_i} q_i - e_3. \quad (7)$$

Using a backward finite difference with the saved previous-frame local strain $\bar{\Gamma}_i$, the damping energy is

$$E_i^d = \frac{c_d}{2h^2} \left| \Gamma_i^{loc} - \bar{\Gamma}_i \right|^2, \quad (8)$$

where h is the time step and $\bar{\Gamma}_i$ is treated as a constant for the purposes of the current solve. We use *stiffness-proportional damping*, setting

$$c_d = \beta k_i^{ss}, \quad (9)$$

where $\beta \geq 0$ is a uniform damping parameter with units of time squared. This is analogous to the stiffness-proportional term in Rayleigh damping: stiffer segments are damped proportionally more, preventing the damping model from artificially softening stiff regions or over-damping soft ones.

Reformulation as a world-space norm. Since quaternion rotation preserves norms, $|\bar{q} \mathbf{a} q| = |\mathbf{a}|$, we rotate the argument of the norm by q_i :

$$|\bar{q}_i \mathbf{p} q_i - (e_3 + \bar{\Gamma}_i)|^2 = |\mathbf{p} - q_i(e_3 + \bar{\Gamma}_i)\bar{q}_i|^2, \quad (10)$$

where $\mathbf{p} = (\mathbf{x}_{i+1} - \mathbf{x}_i)/l_i$. Defining the *damping rest direction*

$$\mathbf{r}_i = e_3 + \bar{\Gamma}_i, \quad (11)$$

the damping energy takes the form

$$E_i^d = \frac{\beta k_i^{ss}}{2h^2} \left| \frac{\mathbf{x}_{i+1} - \mathbf{x}_i}{l_i} - q_i \mathbf{r}_i \bar{q}_i \right|^2. \quad (12)$$

This is structurally identical to the stretching/shearing energy E_i^{ss} with the replacements $e_3 \rightarrow \mathbf{r}_i$ and $k_i^{ss} \rightarrow \alpha k_i^{ss}$, where $\alpha = \beta/h^2$.

3 Torque Derivation

The torque on q_i from the damping energy follows by the same quaternion calculus as for E^{ss} . For an energy of the form $E = \frac{k}{2} |\mathbf{p} - q \mathbf{a} \bar{q}|^2$, the quaternion gradient is

$$-\nabla_q E = -2k (\mathbf{p} - q \mathbf{a} \bar{q}) q \mathbf{a}. \quad (13)$$

Applying this to eq. (12):

$$\tau_{i,i}^d = -\frac{2\beta k_i^{ss}}{h^2} \left(\frac{\mathbf{x}_{i+1} - \mathbf{x}_i}{l_i} - q_i \mathbf{r}_i \bar{q}_i \right) q_i \mathbf{r}_i. \quad (14)$$

Expansion and absorption into λ . Expanding the quaternion product and using $\mathbf{r}_i^2 = -|\mathbf{r}_i|^2$ for pure imaginary quaternions:

$$\tau_{i,i}^d = -\frac{2\beta k_i^{ss}}{h^2} (\mathbf{p} q_i \mathbf{r}_i + |\mathbf{r}_i|^2 q_i). \quad (15)$$

The term $|\mathbf{r}_i|^2 q_i$ is proportional to q_i and is absorbed into the Lagrange multiplier λ , exactly as the analogous term from τ^{ss} is absorbed in the original derivation. The remaining contribution $\mathbf{p} q_i \mathbf{r}_i$ enters the equilibrium equation.

4 Modified Equilibrium

Adding the damping torque to the equilibrium, the $\mathbf{p} q_i \mathbf{r}_i$ contribution becomes:

$$\mathbf{v} q_i e_3 + \mathbf{v}_d q_i \mathbf{r}_i + \mathbf{b} - \lambda q_i = 0, \quad (16)$$

where

$$\mathbf{v}_d = -\frac{2\beta k_i^{ss}}{h^2 l_i} (\mathbf{x}_{i+1} - \mathbf{x}_i) = \alpha \mathbf{v}. \quad (17)$$

Since $\mathbf{v}$ and $\mathbf{v}_d$ are both proportional to $(\mathbf{x}_{i+1} - \mathbf{x}_i)$, we can write $\mathbf{v}_d = \alpha \mathbf{v}$ with

$$\alpha = \frac{c_d}{k_i^{ss} h^2} = \frac{\beta}{h^2}. \quad (18)$$

Note that under stiffness-proportional damping, α is *uniform across all segments*—it depends only on the global parameter β and the time step, not on per-segment stiffness. Factoring:

$$\mathbf{v} q_i (e_3 + \alpha \mathbf{r}_i) + \mathbf{b} - \lambda q_i = 0. \quad (19)$$

Define the *generalized material axis*

$$\boldsymbol{\eta}_i = e_3 + \alpha \mathbf{r}_i = (1 + \alpha) e_3 + \alpha \bar{\Gamma}_i. \quad (20)$$

The modified equilibrium is then

$$\boxed{\mathbf{v} q_i \boldsymbol{\eta} + \mathbf{b} - \lambda q_i = 0.} \quad (21)$$

This has the identical structure to eq. (4) with $e_3 \rightarrow \boldsymbol{\eta}$.

5 Closed-Form Solution

5.1 Derivation of $q(\lambda)$

Right-multiply eq. (21) by $\boldsymbol{\eta}$ and use $\boldsymbol{\eta}^2 = -|\boldsymbol{\eta}|^2$:

$$-|\boldsymbol{\eta}|^2 \mathbf{v} q_i + \mathbf{b} \boldsymbol{\eta} - \lambda q_i \boldsymbol{\eta} = 0. \quad (22)$$

Left-multiply eq. (22) by $\mathbf{v}$ and use $\mathbf{v}^2 = -|\mathbf{v}|^2$:

$$|\boldsymbol{\eta}|^2 |\mathbf{v}|^2 q_i + \mathbf{v} \mathbf{b} \boldsymbol{\eta} - \lambda \mathbf{v} q_i \boldsymbol{\eta} = 0. \quad (23)$$

From eq. (21), rearranging gives $\mathbf{v} q_i \boldsymbol{\eta} = \lambda q_i - \mathbf{b}$. Substituting into eq. (23):

$$|\boldsymbol{\eta}|^2 |\mathbf{v}|^2 q_i + \mathbf{v} \mathbf{b} \boldsymbol{\eta} - \lambda (\lambda q_i - \mathbf{b}) = 0. \quad (24)$$

Collecting the q_i terms and solving:

$$\boxed{q_i(\lambda) = \frac{\mathbf{v} \mathbf{b} \boldsymbol{\eta} + \lambda \mathbf{b}}{\lambda^2 - |\mathbf{v}|^2 |\boldsymbol{\eta}|^2}.} \quad (25)$$

This is the generalized form of eq. (6). When $\boldsymbol{\eta} = e_3$ and $|\boldsymbol{\eta}| = 1$, it reduces to the original.

5.2 Approximate λ

The unit-norm constraint $|q_i(\lambda)| = 1$ and the triangle inequality give bounds

$$|\mathbf{v}||\boldsymbol{\eta}| < \lambda \leq |\mathbf{v}||\boldsymbol{\eta}| + |\mathbf{b}|, \quad (26)$$

and the approximate solution is the upper bound:

$$\boxed{\lambda \approx |\mathbf{v}||\boldsymbol{\eta}| + |\mathbf{b}|}. \quad (27)$$

5.3 Exact λ via fixed-point iteration

The fixed-point operator generalizes directly:

$$\lambda_{fp} = f(\lambda) = \sqrt{|\mathbf{v}\mathbf{b}\boldsymbol{\eta} + \lambda\mathbf{b}| + |\mathbf{v}|^2|\boldsymbol{\eta}|^2}, \quad (28)$$

with γ_i stored such that $\lambda_i = |\mathbf{v}_i||\boldsymbol{\eta}_i| + \gamma_i|\mathbf{b}_i|$.

6 Well-Definedness

Zero strain, steady state. Under zero strain ($C^{ss} = 0$, $C^{bt} = 0$) and steady state ($\bar{\Gamma} = 0$), we have $\mathbf{r} = e_3$ and $\boldsymbol{\eta} = (1 + \alpha)e_3$. Substituting $\mathbf{v} = -|\mathbf{v}|q e_3 \bar{q}$ and $\mathbf{b} = |\mathbf{b}|q$:

$$q(\lambda) = \frac{|\mathbf{b}|(|\mathbf{v}|(1 + \alpha) + \lambda)}{\lambda^2 - |\mathbf{v}|^2(1 + \alpha)^2} q. \quad (29)$$

With $\lambda = |\mathbf{v}|(1 + \alpha) + |\mathbf{b}|$, the prefactor reduces to 1 and $q(\lambda) = q$. The model introduces no artificial strain. ✓

Frame indifference. Rotating all quantities by an arbitrary quaternion R : $\mathbf{v} \rightarrow R\mathbf{v}\bar{R}$, $\mathbf{b} \rightarrow R\mathbf{b}$, $\boldsymbol{\eta}$ is unchanged (it lives in local space), and $|\mathbf{v}|$, $|\mathbf{b}|$, $|\boldsymbol{\eta}|$ are all invariant. The solution $q_i(\lambda)$ transforms as $q \rightarrow Rq$. ✓

7 Position Forces and Hessian

The damping energy also contributes to the VBD position update. The force on vertex $\mathbf{x}_i$ from the damping energy is

$$\mathbf{f}_{i,i}^d = -\nabla_{\mathbf{x}_i} E_i^d = \frac{\beta k_i^{ss}}{h^2 l_i} \left(\frac{\mathbf{x}_{i+1} - \mathbf{x}_i}{l_i} - q_i \mathbf{r}_i \bar{q}_i \right), \quad (30)$$

with diagonal Hessian

$$\nabla_{\mathbf{x}_i} \mathbf{f}_{i,i}^d = \frac{\beta k_i^{ss}}{h^2 l_i^2} \mathbf{I}. \quad (31)$$

These add directly to the existing E^{ss} force and Hessian contributions in the VBD block-diagonal solve.

8 Implementation Summary

The changes to Algorithm 2 of [1] are:

Start of each frame (before the iterative solve): Compute the local strain from the current (i.e. previous frame’s converged) state.

$$\bar{\Gamma}_i \leftarrow \bar{q}_i \frac{\mathbf{x}_{i+1} - \mathbf{x}_i}{l_i} q_i - e_3.$$

Orientation update (replacing the inner loop of Algorithm 2):

$$\begin{aligned} \mathbf{v} &\leftarrow -2k_i^{ss} l_i^{-1} (\mathbf{x}_{i+1} - \mathbf{x}_i) && \text{(unchanged, eq. (5))} \\ \mathbf{b} &\leftarrow \sum \text{bending terms} && \text{(unchanged, eq. (5))} \\ \mathbf{r} &\leftarrow e_3 + \bar{\Gamma}_i && \text{(damping rest direction, eq. (11))} \\ \alpha &\leftarrow \beta/h^2 && \text{(uniform damping ratio, eq. (18))} \\ \boldsymbol{\eta} &\leftarrow (1 + \alpha) e_3 + \alpha \bar{\Gamma}_i && \text{(generalized axis, eq. (20))} \\ \lambda &\leftarrow |\mathbf{v}| |\boldsymbol{\eta}| + |\mathbf{b}| && \text{(approximate, eq. (27))} \\ q_i &\leftarrow \mathbf{v} \mathbf{b} \boldsymbol{\eta} + \lambda \mathbf{b} && \text{(generalized solve, eq. (25))} \\ q_i &\leftarrow q_i / |q_i| \end{aligned}$$

Position update: Add the damping force (eq. (30)) and Hessian (eq. (31)) contributions to the existing stretching/shearing terms in the VBD block-diagonal solve.

The solver structure, parallelism, and convergence properties are all preserved. No additional persistent per-segment storage is required— $\bar{\Gamma}_i$ is computed once at the start of each frame and reused across iterations.

References

- [1] Jerry Hsu, Tongtong Wang, Kui Wu, and Cem Yuksel. Stable Cosserat Rods. In *SIGGRAPH Conference Papers '25*, August 2025. doi:10.1145/3721238.3730618.